Principal Directions versus Principal Subspaces

Spectral Stability of Equity Covariance

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Abstract

A principal direction is not always uniquely determined by a covariance matrix. We examine eigenvector nonuniqueness arising from repeated eigenvalues and sign choices, and distinguish it from the uniqueness of an invariant principal subspace. A three-dimensional example shows that rotating an eigenbasis inside a repeated-eigenvalue subspace leaves its orthogonal projector unchanged. Finally, a clustered spectrum illustrates why the separation of a selected subspace from its complement is more relevant to subspace stability than gaps between eigenvalues within that subspace.

1 Repeated eigenvalues and principal directions

Let $\Sigma \in \mathbb{R}^{d\times d}$ be a real symmetric positive semidefinite covariance matrix. A unit vector v is an eigenvector associated with λ if $\Sigma v = \lambda v$; its associated eigenspace is

$$E_\lambda = \ker(\Sigma - \lambda I_d).$$

The distinction between an eigenvector and its eigenspace is essential when an eigenvalue is repeated.

Proposition 1.1 (Nonuniqueness under complete degeneracy)**.** *For $\Sigma = 5I_2$, every unit vector $v \in \mathbb{R}^2$ is an eigenvector with eigenvalue 5, and every unit vector maximizes the Rayleigh quotient $v^\mathsf{T}\Sigma v$.*

Proof. For any $\theta \in [0, 2\pi)$, let

$$v(\theta) = \begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}.$$

Then

$$\Sigma v(\theta) = 5v(\theta), \qquad v(\theta)^\mathsf{T}\Sigma v(\theta) = 5(\cos^2\theta + \sin^2\theta) = 5.$$

Thus neither an individual maximizing direction nor its one-dimensional span is uniquely determined. Nevertheless, the full eigenspace $E_5 = \mathbb{R}^2$ is uniquely determined by Σ. □

For example, the orthonormal bases

$$\left\{\begin{pmatrix}0\\1\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}\right\} \quad \text{and} \quad \left\{\frac{1}{\sqrt{2}}\begin{pmatrix}-1\\1\end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}\right\}$$

are different choices of eigenvectors for the same eigenspace.

2 Rotations within a leading eigenspace

Consider the three-dimensional covariance matrix

$$\Sigma = \begin{pmatrix}6 & 0 & 0\\ 0 & 6 & 0\\ 0 & 0 & 1\end{pmatrix}, \qquad \lambda_1 = \lambda_2 = 6, \quad \lambda_3 = 1. \tag{1}$$

Although the first two eigenvalues coincide, their common eigenspace is well defined:

$$E_6 = \ker(\Sigma - 6I_3) = \operatorname{span}\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}. \tag{2}$$

Its dimension is two, and the boundary eigengap is $\lambda_2 - \lambda_3 = 5 > 0$.

Proposition 2.1 (Different eigenbases, one projector)**.** *Let*

$$V_1 = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & 0 \end{pmatrix}, \qquad V_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Both matrices have orthonormal columns spanning E_6, and their orthogonal projectors are equal:

$$\boxed{V_1 V_1^\mathsf{T} = V_2 V_2^\mathsf{T} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.} \tag{3}$$

Proof. Every column of V_1 or V_2 has third coordinate zero, so it lies in E_6. The two columns of each matrix are orthonormal. Direct multiplication gives

$$V_1 V_1^\mathsf{T} = \begin{pmatrix} \frac{3}{4} + \frac{1}{4} & \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} & 0 \\ \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} & \frac{1}{4} + \frac{3}{4} & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

which also equals $V_2 V_2^\mathsf{T}$. □

Thus the two eigenbases contain different individual principal directions, but describe the same leading two-dimensional principal subspace.

3 Sign ambiguity and spectral clusters

Proposition 3.1 (Eigenvector sign ambiguity)**.** *If $\Sigma v = \lambda v$ and $\|v\|_2 = 1$, then $-v$ is another unit eigenvector for λ. Although*

$$\|v - (-v)\|_2 = 2,$$

the associated rank-one orthogonal projectors agree:

$$\boxed{vv^\mathsf{T} = (-v)(-v)^\mathsf{T}.} \tag{4}$$

Proof. By linearity, $\Sigma(-v) = -\Sigma v = \lambda(-v)$. Moreover, $\|v - (-v)\|_2 = \|2v\|_2 = 2$, while $(-v)(-v)^\mathsf{T} = (-1)^2 vv^\mathsf{T} = vv^\mathsf{T}$. □

A further distinction arises for clusters of nearby eigenvalues. For example, consider

$$\Sigma = \mathrm{diag}(10, 9.9, 9.8, 2, 1).$$

The consecutive eigengaps are

$$\lambda_1 - \lambda_2 = 0.1, \qquad \lambda_2 - \lambda_3 = 0.1, \qquad \lambda_3 - \lambda_4 = 7.8. \tag{5}$$

The first three eigenvalues form a closely spaced group, separated from the remaining eigenvalues by a substantially larger gap. This distinction motivates comparing the *top-three subspace* rather than tracking individual labeled eigenvectors inside the cluster. The gap pattern alone does not quantify perturbation error; a stability bound also depends on the perturbation magnitude and suitable separation assumptions.

4 Basis invariance of principal subspaces

The preceding examples are instances of a general algebraic identity.

Proposition 4.1 (Invariance under orthogonal change of basis)**.** *Let $V_1 \in \mathbb{R}^{d\times k}$ have orthonormal columns, and let $V_2 = V_1 Q$, where $Q \in \mathbb{R}^{k\times k}$ is orthogonal ($Q^\mathsf{T}Q = QQ^\mathsf{T} = I_k$). Then V_1 and V_2 span the same subspace and determine the same orthogonal projector.*

Proof. Since Q is invertible, $\mathrm{im}(V_1 Q) = \mathrm{im}(V_1)$. Furthermore,

$$V_2 V_2^\mathsf{T} = V_1 Q Q^\mathsf{T} V_1^\mathsf{T} = V_1 V_1^\mathsf{T}. \tag{6}$$

□

Therefore individual principal directions may change with a choice of sign or orthonormal basis, even when the underlying principal subspace does not change. For covariance-stability analysis, the orthogonal projector $P = VV^\mathsf{T}$ provides a basis-independent representation of that subspace.